

ECONOMIC ZOOLOGY (PAST PAPER)

PAST PAPER (2014)

Q#1 How to use Archaeopteryx for Research purpose?

Archaeopteryx is a primitive bird that is often used as an example of how fossils can show transitional features of evolution. Archaeopteryx is a fossil species of dinosaur that is classified as a Coelurosaurian dinosaur. Coelurosaurian dinosaurs include theropod dinosaur species that gave rise to modern birds. Archaeopteryx is believed to be the first dinosaur to evolve with wings. It had broad wings with rounded ends and was about the size of a raven. Archaeopteryx is considered to be the missing link between reptiles and birds in the evolution process. It has traits of both dinosaurs and birds, including feathers and a beak, which are unique to birds, and teeth and a long bony tail, which are traits of dinosaurs.

Here are some other facts about Archaeopteryx:

- It existed from 152.1 million years ago to Barremian Age
- It lived in a coastal land
- It was an insectivore
- It reproduced by laying eggs
- It could probably fly, but it probably wasn't a strong flyer
- It was a small carnivore that fed on insects, lizard-like reptiles, and other small organisms
- It had a full set of teeth, a long bony tail, and three claws on its wing

Q#2 Name the species of Honey Bee that can be domesticated?

Only two species have been truly domesticated: *Apis mellifera* and *Apis cerana*. *A. mellifera* has been cultivated at least since the time of the building of the Egyptian pyramids, and only that species has been moved extensively beyond its native range

Apis mellifera

Also known as the western honeybee or European domestic bee, this is the most commonly domesticated species of honeybee. It is the main species that has been domesticated by humans, largely due to its habit of building its nests within cavities.

Apis cerana indica

Also known as the Indian hive bee or Asian bee, this species is native to India and Asia. They are the domesticated species that build multiple combs, with an average yield of 6-8 kg/year

Q#3 What happens to the Drones after mating flight?

After mating, male honey bees (drones) die quickly. The drone's endophallus is removed from his body and attached to the queen during mating. The drone attempts to fly away, but the endophallus remains in the queen, causing the lower half of his abdomen to rip open and he dies.

Drones are only able to mate seven to 10 times during a mating flight. Even drones that survive the mating flight are ejected from their nests.

During mating season, drones fly high in the air and gather together in Drone Congregation Areas (DCA's). They use their giant eyes and scent receptors to lure and identify a virgin queen.

Q#4 Define Piggery?

A piggery is a farm or building where pigs are kept. It can also refer to a workplace made up of farm buildings and cultivated land

Synonyms of piggery include:

Pigsty, Stockyard, Pigpen, Enclosure, Pen, Sty, Hogpen.

Q#5 Why are pesticides considered to be hazardous?

Pesticides are considered hazardous because they can be toxic to humans and other organisms. Pesticides can also contaminate soil, water, turf, and other vegetation.

Pesticides can cause a variety of health issues, including:

- Nervous system: Some pesticides, such as organophosphates and carbamates, can affect the nervous system
- Skin or eyes: Some pesticides can irritate the skin or eyes
- Carcinogens: Some pesticides may be carcinogens

Hormone or endocrine system: Some pesticides may affect the hormone or endocrine system in the body

Q#6 What is difference b/w Insects & Pests?

A **pest** is a living organism that is undesirable due to a negative trait or behavior. Pests can be vertebrates, including rodents and birds, or invertebrates, such as insects and even some harmful plants.

Insects are small air-breathing arthropods with a body divided into head, thorax, and abdomen. They have six legs, a pair of antennae, and zero or two pairs of wings.

Not all insects are pests. Many different kinds of insects eat other insects and are beneficial species. Some insects are beneficial for plants.

Examples of pests include:

Mosquitoes, Rodents, Weeds, Ants, bees, and wasps, Bed bugs, Cockroaches, Fleas, Flies, Myriapods, Pigeons.

Q#7 What is the best way to control cestode parasite problems?

Here are some ways to control cestode parasite problems:

- **Drugs:** Albendazole and praziquantel are the primary drugs used to treat cestode infections. Albendazole prevents the helminth from absorbing glucose, which inhibits the production of energy. Praziquantel causes the worm's musculature to contract in a tetanus-like manner. Praziquantel is also the most commonly used medicine for tapeworms.

- **Hygiene:** Good hygiene can help prevent the spread of tapeworms.
- **Cooking:** Meat should be cooked thoroughly or frozen at -10°C for 10 days. Beef and pork should be inspected for *Taenia*, also known as "measly meat".
- **Monitoring:** Monitoring dog and fox populations can help anticipate risks for human exposure.
- **Corticosteroids:** Corticosteroids can reduce swelling and other immune system activity.

Q#8 What is the Economic Importance of Fishes?

Fish are economically important for a number of reasons, including:

➤ **Food**

Fish are a valuable source of protein, vitamins, and minerals. They are a staple food in many developing countries and a healthy luxury food in the rich world. Fish are also low in sugar and can be part of a healthy diet for diabetics.

➤ **Jobs**

Fish provide jobs for fishers, traders, and dealers. The fishing industry contributes to a country's GDP and supports the livelihoods of coastal communities.

➤ **By-products**

Fish produce a number of commercially valuable by-products, including fish oil and fish meal. Fish oil is used in soaps, paints, and cosmetics. Fish meal is made from fish waste and used as food for pigs, poultry, and cattle.

Exports

Fish exports are a significant source of foreign exchange for developing countries. For example, in 2021-2022, Bangladesh generated \$533 million from the export of fish, shrimp, and other fishery products.

Fish also have medicinal value. Fish insulin can be used when a lower concentration of insulin is needed for treatment. Fish liver oil is rich in vitamin A and P and has medicinal value.

Q#9 What is Pearl Harvesting?

Pearl harvesting, also known as pearling, is the process of recovering pearls from mollusks, such as oysters or mussels, in freshwater or the sea.

The process of harvesting pearls is as follows:

1. Remove the oysters from the water
 2. Transport the oysters to a harvesting facility
 3. Have a harvester carefully open the oyster
 4. Remove the pearl using a surgical instrument
 5. Prepare the oysters to create more pearls
- Removing a pearl does not harm the oyster. In fact, the oyster can then be renucleated to produce another pearl in 18 months to 2 years.

Q#10 What is the Economic Importance of Rats and their control?

Rats have a number of economic benefits and drawbacks:

Benefits:

- **Scientific and pharmaceutical testing:** Rats have been used in research on cardiovascular medicine, wound healing, diabetes, transplantation, behavioral studies, and space motion sickness. They have also been used to test the safety and efficacy of drugs.
- **Food and fur:** Some rodent species provide food and fur for human use.
- **Seed and spore dispersal:** Rodents are important in seed and spore dispersal, pollination, and energy and nutrient cycling.
- **Food source for predators:** Rodents are a food source for many predators

Some strategies for controlling rats include:

- Eliminating food, water, and cover/climbing sources
- Storing pet food in rodent proof containers
- Not leaving excess food in pet dishes for extended periods of time
- Gathering ripe fruit in your yard and surrounding areas

Drying up water sources when possible

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Q#1 Write down the three problem of Pearl Industry?

BIOFOULING AND PREDATION. Major problems in pearl oysters farming are caused by biofouling organisms which settle and grow on the oyster shells, by boring organisms which riddle through the shells making them weak and friable, and by predators which feed on the oysters.

Q#2 Enumerate methods of preservation of Fishes?

Here are some methods of fish preservation:

- Freezing
Helps retain the fish's natural appearance. Freezing methods include blast freezing, plate freezing, immersion freezing, and spray freezing.
- Smoking
Exposes fish to smoke from smouldering wood or plant materials to add flavor and preservative ingredients. Smoking can also include salting, drying, and heating steps.
- Drying
The oldest known preservation method, dried fish can last for several years.

➤ Salting

A common practice in India and other parts of the world, salting is a partial dehydration of fish by osmosis with sodium chloride.

➤ Smoking drying

Removes moisture and prevents bacteria growth.

➤ Formalin

A common preservative for fish specimens, formalin is typically injected into the fish or prawn.

Other methods of fish preservation include:

- Canning
- Pickling
- Chilling
- Curing
- Marinating
- Fermentation
- Modified atmospheric packing
- Retort pouch packaging

Q#3 What are two important diseases of Poultry?

Here are some common diseases:

• **Chronic diseases**

These include heart disease, cancer, and diabetes, and are the leading causes of death and disability in the United States.

Infectious diseases

These are illnesses caused by harmful agents (pathogens) that get into your body. The most common causes are viruses, bacteria, fungi, and parasites. Examples of infectious diseases include strep throat, urinary tract infections, and tuberculosis.

Other diseases that affect poultry include:

Infectious bronchitis, Newcastle disease, Infectious laryngotracheitis, Pneumovirus, Air sac syndrome, Infectious sinusitis.

This is for informational purposes only. This information does not constitute medical advice or diagnosis.

Q#4 The causative agent of Pyorrhoea of Riggs disease?

Riggs' disease, also known as pyorrhea of a toothsocket, is a historical term for periodontitis (gum disease). Pyorrhea is bacterial in nature and is the result of advanced or untreated gingivitis. The most common type of pyorrhea is chronic pyorrhea, which is caused by gradual plaque buildup. Plaque is a sticky film of bacteria that builds up on the teeth and hardens. When bacteria stay on the teeth long enough, they form a film called plaque, which eventually hardens to tartar, also called calculus.

Other factors that may place an individual at risk for developing disease include: Tobacco smoking, Diabetes.

Q#5 Differentiate b/w Stifling and Reeling?

Inside the cocoons, the larvae undergo metamorphosis and turn into pupae. The harvesting of silk from these cocoons is the final stage of sericulture. First, the pupae inside the cocoon are killed by boiling the cocoon and exposing it to steam and dry heat. This process is called **stifling**.

Now, the silk filaments are removed from the dead cocoon via a process called **reeling**. When the cocoons are placed in boiling water for approximately 15 minutes, the adhesion of the silk threads reduces, enabling the separation of individual filaments. These filaments are twisted into a thread with the help of a series of guides and pulleys. This silk is then re-boiled in order to improve its **lustre**

Q#6 Direct value of Mammals?

Mammals provide many direct and indirect benefits to humans, including:

- **Ecosystem services:** Mammals play a vital role in ecosystems by pollinating, dispersing seeds, regulating insect populations, and reducing disease transmission. Some mammals may also serve as indicators of ecosystem health.
- **Food:** Mammals provide dairy products and meat, and their hides and wool can be used for clothing and equipment.
- **Transport and power:** Until the advent of mechanized transport, domesticated mammals provided much of the power used for work and transport.
- **Medicines:** Many mammals are used in traditional health care.
- **Shelter:** Mammals can provide shelter.

Pakistan has 195 mammal species, including 13 subspecies. The livestock sector contributes about 56.3% of the value of agriculture and nearly 11% to the agricultural gross domestic product (AGDP). Milk is the most important commodity in the Pakistani livestock sector.

Q#7 Appliances for Apiculture?

Apiculture, also known as beekeeping, is the commercial production of honey and other products. Here are some basic tools and equipment for apiculture:

- **Hives**
Man-made structures where honey bee colonies live. Hives can be movable and made of a wooden box.
- **Brood chambers**
Wooden boxes that contain honeybees raised in brood chambers. Brood chambers have a wooden platform with an opening at the bottom for the bees to enter and exit.
- **Hive frames**
Structural elements that hold the honeycomb or brood comb within the hive. Hive frames can be removed to inspect the bees or extract honey.

- **Bee brush**
A 14-inch long tool with soft yellow bristles that can be used to gently remove bees from frames, honey supers, or other areas.
- **Hive tool**
A tool that can pry apart hive boxes and frames that have been glued together with bee propolis.
- **Plastic queen gate**
A barrier placed at the entrance of the bee hive to prevent the queen from leaving.
- **Bee veil**
A net or veil that can be worn over the head to protect the face from bee attacks while removing honey from a hive.

Q#8 Prevention and control of Cockroach?

Here are some tips to prevent and control cockroaches:

- **Clean** Cockroaches are attracted to crumbs and spills, so it's important to clean up spills and sweep floors regularly. You should also clean up dirty dishes and food or drink spills as soon as possible.
- **Store food properly** Make sure all food waste is placed in tightly sealed bags and disposed of quickly.
- **Seal entry points** Cockroaches can squeeze through small crevices and cracks, so seal gaps in windows, doors, and walls.
- **Fix leaky pipes** Cockroaches like damp areas, so fixing leaky faucets and pipes is vital in infestation prevention.
- **Block entry points** Cockroaches can never come into your home if you stop them in the first place.
- **Use bait** Cockroach bait is one of the most common methods of cockroach control.
- **Use boric acid** Boric acid is a wonderful tool for controlling cockroaches in homes, restaurants and other buildings.
Hire a professional pest control company A professional pest control company will take care of eliminating a cockroach fast, and can offer a guarantee that no new cockroach will infest your home within an agreed upon period of time.

Q#9 Two important pests of Stored Grains?

Indianmeal moth

This moth is the most economically important pest of stored grains. All the larval stages of this moth feed on the germs of different kinds of cereal and rice grains. They also feed on processed dry foods like pasta, bread, and spices.

- **Grain weevils**
These are important pests of farm-stored grain. They can infest undamaged grain and also attack other hard cereal products, like macaroni and spaghetti.
- **Rice weevil**
This is a primary pest of stored grain. Its grub-like larva develops in a sound kernel.

Q#10 Pests of Tea

Tea plants can be attacked by many pests, including insects, mites, nematodes, diseases, and weeds.

Here are some pests that can affect tea:

- **Red spider mites**
A major pest in most tea-producing countries. Mites are a serious dry weather pest that can damage the green tissue of leaves.
- **Thrips**
Both nymphs and adults are destructive, with nymphs causing more harm. Thrips complete their life cycle in 14-20 days and suck the cell sap, leaving two longitudinal sandpaper-like lines on the leaves.
- **Shot hole**
A major stem pest.
- **Aphids**
Toxoptera aurantii Boyer de Fonscolombe (Aphididae: Hemiptera).
- **Tea mosquito bug**
Helopeltis theivora Waterhouse (Miridae: Hemiptera).

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Q#1 Briefly describe any two diseases of poultry?

Here are two diseases that affect poultry:

Pullorum disease

A bacterial disease that affects chickens and turkeys. It's usually asymptomatic in adults, but is transmitted to young chicks through eggs.

Avian influenza (AI)

A highly contagious respiratory viral disease that affects both domestic and wild birds. AI viruses are classified into two pathotypes: low pathogenic avian influenza (LPAI) and highly pathogenic avian influenza (HPAI). LPAI typically causes few to no clinical signs in poultry. HPAI is a very infectious and fatal form of the disease.

Other diseases that affect poultry include:

Fowl typhoid, Salmonella infection (paratyphoid and pullorum), Colibacillosis, Infectious coryza, Ornithobacterium rhinotracheale, Riemerella anatipestifer, Salmonellosis, Staphylococcal infection, Streptococcal infection.

Q#2 Write a Brief note on Pearl formation?

Pearls are made by marine oysters and freshwater mussels as a natural defence against an irritant such as a parasite entering their shell or damage to their fragile body. The oyster or mussel slowly secretes layers of aragonite and conchiolin, materials that also make up its shell.

Q#3 Write down the properties of Lac?

Lac resin is thought to be a polyester complex of straight chain hydroxy fatty acid of C14-C18 carbon

chain (e.g., Aleuritic acid, butolic acid), mono- and di-hydroxy acid. Other components include dye (1-2%), wax (6%), sugar, protein, soluble salts, sand, woody matter, insect body debris, etc.

Lac is used in the manufacture of toys, ornaments, electrical insulating materials, varnishes, polishes, lithographic inks, shoe polishes, sealing wax, etc. It is used in electric insulations, lamination of papers, hat proofing and coating of pictures and fossils.

- **Q#4 Differentiate b/w stifling & Realing?**

Question no 05(2016)

- **Q#5 Enumerate methods of preservation of Fishes?**

Question no 01(2016)

- **Q#6 What is Riggs disease?**

Question no 04(2016)

- **Q#7 Write down the control of Ticks on the host?**

Treatment of hosts with acaricides to kill attached larvae, nymphs, and adults of ixodid ticks and larvae of argasid ticks has been the most widely used control method. A variety of ectoparasiticide products are available on the market to treat animals against ticks.

- **Q#8 Pests of tea?**

Question no 10(2016)

- **Q#9 Explain the Morphology & function of Queen Bee?**

Morphology

A queen bee is the largest bee in the hive, measuring around 20 mm in size. She has a long, narrow abdomen with a pointed end, shorter wings, and splayed legs.

Function

A queen bee's primary functions are to lay eggs and produce chemical scents that help regulate the unity of the colony. She is the only female with fully developed ovaries and is usually the mother of most, if not all, of the bees in the hive.

Here are some other details about queen bees:

- They live for about five years.
- They receive sperm from a drone during their nuptial flight and store it in the spermatecae.
- They lay both fertilized and unfertilized eggs.
- All fertilized eggs develop into females.
- The larvae developing from the fertilized eggs are fed with royal jelly for the first four days.
- The other larvae are fed on bee bread (honey and pollen) and become workers (sterile females).

They are developed from larvae selected by worker bees and specially fed in order to become sexually mature.

- **Q#10 Give brief account of indoor and outdoor duties of worker bees?**

Worker bees have different roles throughout their 5–7 week lives. For the first 15 days, they work inside the hive as "house" bees. For the second half of their lives, they work outside the hive as "field" bees.

Here are some of the indoor duties of worker bees:

- Housekeeping: Cleaning, nursing, and building combs
- Feeding: Feeding the queen, drones, and larvae
- Collecting: Collecting pollen and nectar
- Making: Making wax

- **Caring:** Caring for the eggs and larvae
- **Heating and cooling:** Heating and cooling the hive
- **Here are some of the outdoor duties of worker bees:**
- **Foraging:** Gathering nectar, pollen, water, and propolis from the surrounding environment
- **Ripening:** Ripening honey in the honey stomach
- **Repairing:** Using propolis to repair cracks in the hive
- **Maintaining temperature:** Fanning their wings to ventilate and maintain the temperature of the hive

Communicating: Dancing for their sisters to tell them the best places to go

PAST PAPER(2018)

Q#1 Value of Mammals?

Question no 06(2016)

Q#2 Enumerate two by-product of poultry?

Here are two by-products of poultry:

- **Manure:** Has economic value, whether sold or applied to crops.
- **Eggshells:** Used in the food, feed, pharmaceutical, medical, and engineering industries.

Other poultry by-products include:

Hatchery waste, Feathers, Blood, Egg shells, Offal, Fat, Heads, legs, bones, viscera.

Poultry by-products can be processed into livestock feed, fertilizers, pet foods, or discarded.

Q#3 Two important measure of control of House-flies?

Here are some ways to control house flies:

- **Sanitation**
Remove food residue, clean garbage cans weekly, and cover garbage. You can also mix apple cider vinegar and dish soap to attract and kill flies.
- **Eliminate breeding sites**
Remove fly larval sources such as dog manure, garbage, and rotting fruit. You can also remove urine and manure-saturated straw to kill biting stable fly larvae.
- **Use traps**
Place sticky traps and ultraviolet light traps around your home or business. You can also hang resin strips (flypa-per) in infested areas where there is little or no air movement.
- **Use sprays and insecticides**
You can use fly sprays or insecticides to eliminate individual flies. Pyrethrin, a pesticide that is toxic to insects, can be bought as a spray to deal with adult flies.

Use other methods

You can try these other methods to keep flies away:

- Use essential oils
- Seal up windows
- Turn down your heat
- Light citronella candles
- Clean up spills promptly

Q#4 Two recognized forms of malaria caused by different species of plasmodium?

Malaria is caused by five parasite species that are transmitted by the bites of infected female Anopheles mosquitoes. Two of these species are Plasmodium falciparum and Plasmodium vivax.

Here are some differences between these two species:

- **Plasmodium falciparum**

Responsible for more deaths than Plasmodium vivax. It can cause multiple infections of red blood cells (RBCs).

Plasmodium vivax

The most widespread of all malaria species. It can cause severe, even fatal infections. Among patients hospitalized with a primary diagnosis of vivax malaria, about **10–20%** will be classified as having severe illness, and **5%–15%** of those will not survive.

Q#5 Two wild species of Honey Bees?

Here are some wild honey bee species:

- **Apis cerana**

This species has two subspecies, one brown and one dark-brown. It is preferred for domestication because it is less aggressive.

- **Apis dorsata**

Also known as the rock bee, this species is native to Southeast Asia. It builds natural hives in the forest and collects honey without human interference.

- **Apis florea**

Also known as the red dwarf honey bee, this species is small and wild and is found in Southern Asia, such as Cambodia.

Other honey bee species include:

- **Apis mellifera**

This is the scientific name for the honey bee in America.

- **Apis mellifera capensis**

Also known as the Cape bee, this subspecies of the western honey bee can asexually reproduce.

- **Apis nigrocincta**

Also known as the Sulawesi honey bee, this species is highly social and nests in cavities.

Apis laboriosa

This is the only bee that produces red honey, which is known for its medicinal value and relaxing qualities.

Q#6 African Sleeping Sickness?

African sleeping sickness, also known as African **trypanosomiasis**, is a parasitic disease caused by the microscopic parasite **Trypanosoma brucei**. It's transmitted to humans when bitten by an infected tsetse fly, which can only be found in sub-Saharan Africa. The parasites are transmitted from infected humans or animals to the fly's saliva.

Symptoms of African sleeping sickness include:

- Fatigue
- High fever
- Headaches
- Muscle aches
- Irritability
- Swollen lymph nodes
- Aching muscles and joints
- Skin rash

If left untreated, African sleeping sickness can lead to coma and death. Treatment includes taking antiparasitic drugs.

If bitten by a tsetse fly, you should seek immediate medical attention if the bite is painful and symptoms appear.

Q#7 Appliances for Sericulture?

Here are some appliances used in sericulture:

- **Rearing stands:** Made of wood, steel, or bamboo, these stands can hold up to ten trays. The trays can be arranged on shelves, and are used for keeping trays, feeding silkworms, harvesting cocoons, and cleaning dead worms.
- **Mountages:** Used to support larvae when they spin cocoons. Different types of mountages are used in different parts of India.
- **Trays:** Circular trays made of split bamboo can be used to leave worms. Trays can also be used for feeding silkworms, harvesting cocoons, and cleaning dead worms or cocoons.
- **Chopsticks:** Tapering bamboo rods that are 1 cm in diameter can be used to pick younger stages of larvae.
- **Feathers:** Large, white bird feathers can be used to brush newly hatched worms to prevent injuries.

- **Hygrometer:** An instrument used to measure the amount of humidity in the air.
- **Power sprayer:** A device used in sericulture.

Other appliances used in sericulture include:

Bamboo racks, Bamboo baskets, Chopping knife, Foam pads, Wax coated paraffin papers, Nylon nets, Gunny bags, Drier.

Q#8 What is Riggs disease?

Question no 06(2017)

Q#9 Two important Edible Freshwater fishes?

Some important edible freshwater fish include:

- **Trout**
A tasty and nutritious fish that's high in protein and omega-3 fatty acids. Rainbow trout is a good source of lean protein, minerals, and omega-3 fatty acids.
- **Tilapia**
A popular freshwater fish that's eaten in India.

Catfish

A freshwater fish that's considered one of the best to eat.

Other edible freshwater fish include: Perch, Walleye, Largemouth bass, Salmon, Bluegill.

Some other common edible freshwater fish include: **Catla, Rohu, Singhi.**

Q#10 Control of Mosquito?

Here are some ways to control mosquitoes:

1. **Remove standing water**
Mosquitoes breed in standing water, so remove buckets, puddles, wheelbarrows, and bins. You can also change the water in pet drinking bowls, bird baths, and vases once a week.
2. **Use mosquito repellents**
Use repellents that contain picaridin or diethyltoluamide (DEET) on all exposed skin. You can also try using camphor, essential oils like lemon eucalyptus or lavender, or a garlic spray.
3. **Use mosquito traps**
Use traps to determine what species of mosquitoes are present in an area. You can also create a water mosquito trap.
4. **Use larvicides**
Larvicides kill mosquito larvae before they become biting adults. The Environmental Protection Agency (EPA) registers larvicides that can be used by the public and professionals.

5. Use natural predators

Use the natural enemies of mosquitoes to manage their population. This includes introducing parasites, pathogens, and predators to target mosquitoes.

6. Use plant-based bioinsecticides

Four classes of insecticides are commonly used in mosquito control programs:

- a. **Pyrethroids**, such as deltamethrin, permethrin, cypermethrin, and lambda-cyhalothrin
- b. **Organochlorines**, such as DDT
- c. **Organophosphates**, such as malathion and fenitrothion
- d. **Carbamates**, such as propoxur and bendiocarb

PAST PAPER(2019)

Q#1 Write down the problems of sericulture industry?

Some important challenges faced in sericulture are listed below.

- The pebrine disease can infect the eggs, resulting in their death before the hatching of the larvae. Any larvae affected by this disease develop dark spots and become lethargic.
- Viral infections in the larvae may result in the shrinkage of their bodies. They may also start giving off an unpleasant odour.
- Other viral infections such as cytoplasmic polyhedrosis can cause the larvae to lose their appetites.
- The muscardine infection, caused by fungi, can cause the larvae to become extremely feeble and eventually die.
- The larvae of dermestid the beetles notepad
- can bore into the silkworm cocoons and eat the pupae. Silk cannot be reeled from these damaged cocoons.
- Some mites produce a toxic substance that kills silkworms.

Q#2 Enumerate methods of preservation of fishes?

Question no 02(2016)

Q#3 What are two important diseases of poultry?

Question no 01(2017)

Q#4 Two uses of Honey

It has many uses in our day to day life.

- It is used for medicinal purposes.
- It is used on burns.
- It is used to make food.
- It also helps to improve memory.

Q#5 Two important pests of Household-Goods?

Here are some common household pests:

- Cockroaches and ants
Can contaminate food and spoil surfaces
- Termites
Can break down wood and damage furniture, cardboard boxes, paper products, and more
- Bed bugs
Can hitchhike on clothing and luggage and can be devastating to a home
- Flies
Houseflies and fruit flies are two common household pests
- Spiders
Can create webs and feed on insects
- Mosquitoes
Can transmit diseases to humans and animals
- Mice
Can transmit disease, spoil food, and destroy property

Fleas

Can bite discomfortingly and transmit diseases such as typhus and plague

Q#6 Three casts of honey Bees

The three castes of honey bees are:

- **Drones**
Male honey bees with larger heads and thoraxes than females. Drones mate with queen bees.
- **Workers**
Usually non-reproductive females. Worker bees build nests, collect food, and raise brood. Young worker bees, called nurse bees, feed larvae, process nectar, and feed the queen. Older worker bees leave the hive to gather resources for the colony.

Queens

Reproductive females that produce eggs. Queens are longer and more slender than drones, but their faces and thoraxes look similar to workers.

A honey bee colony is made up of thousands of worker bees and one queen. Each member of the colony has a specific task based on its age.

Q#7 Two products of Apiculture?

Apiculture, also known as beekeeping, is the scientific rearing of honeybees for the commercial production of honey and other beneficial products.

Two primary products of beekeeping are honey and wax. Other marketable primary bee products include:

Pollen, Propolis, Royal jelly, Venom, Queens, Bees, Their larvae.

Honey and beeswax are the major products of apiculture. Honey is of high medicinal and nutritional value, and beeswax is used for making candles.

These natural substances have several uses, for both bees and humans, as foods, sealants, lubricants, construction material, and medications.

Q#8 Prevention & Control of Rats

Here are some ways to prevent and control rats:

- **Remove food sources**
Remove food sources, water, and shelter for rats. Rats need access to water daily, between 15ml and 60ml.
- **Remove nesting areas**
Prune shrubs, vines, and trees away from buildings, fences, and other trees. Stack piles of lumber, boxes, etc., at least 18 inches above ground and 12 inches from walls.
- **Use repellents**
Use essential oils such as peppermint oil and other smells that rats don't like. Other natural rat deterrents include crushed pepper, cayenne pepper, black pepper, cloves, citronella, eucalyptus, and chilli flakes.
- **Seal entry points**
Seal all holes and cracks in foundations, walls, floors, underneath doors and around windows.
- **Use baits and poisons**
Use rodenticides, which include grains, seeds, nuts and other foods that contain chemicals lethal to mammals. Rodenticides are usually employed when other methods of eliminating rat populations have failed or when the population is especially large.
- **Use other approaches**
Use dry ice, set traps, use fumigants, or try repellents.

Q#9 Two important pest of Stored-Grains?

Question no 09(2016)

Q#10 Two pests of Fruits?

Here are some pests that can affect fruit:

- **Aphids**
These tiny insects are also known as greenfly and blackfly. They can weaken plants by sucking their sap and transmitting plant viruses.

- **Japanese beetles**

The larvae of Japanese beetles can damage lawns, and the adults can skeletonize trees and shrubs. They can also attack flower buds and fruit.

- **Spider mites**

These mites are members of the arachnid class, along with spiders and ticks. They feed on many fruit trees, veggies, and other plants.

- **Thrips**

These small insects are an economic pest of pome and stone fruit. Damage can occur at flowering (early season) and/or when the crop matures (late season).

- **Fruit flies**

These pests are one of the most destructive pests of fruits and vegetables around the world.

- **Scales**

These pests destroy plant branches and fruits by sucking on the plant host's sap. They look like odd bumps on plant stems, leaves, or fruit.

- **Bark borer**

The bark over infested areas often is dead and dark-colored. This borer can be very destructive.

Codling moth

This pest destroys millions of pounds of fruit. They are major pests to crops, fruits such as apples and pears.

PAST PAPER(2020)

Q#1 Define piggery?

Question no 01(2014)

Q#2 What are dagger nematodes?

Dagger nematodes, also known as *Xiphinema spp.*, are ectoparasitic nematodes that feed on the roots of host plants. They are important crop pests in many North American states and are widely found in tropical countries.

Dagger nematodes are migratory, so they are rarely observed feeding on host plants. They are ectoparasites, which means that their bodies remain in the soil while their odontostyle penetrates the root cortex to feed. The nematode inserts its long stylet deep into the root while the body remains outside the root, in the soil. The stylet punctures cell walls as it penetrates plant tissues.

Dagger nematodes feed on epidermal cells near the root tip. Concentrated feeding may initiate root bending, followed by lesion darkening. Extensive attack causes root death and induces the production of

tufts of lateral roots.

Q#3 Differentiate b/w mites & insects?

Mites are not insects, but are related to spiders. Here are some differences between mites and insects:

- Body parts: Mites have two body regions, the cephalothorax and abdomen. Insects have three major body parts, the head, thorax, and abdomen.
- Legs: Mites can have two, three, or four pairs of legs, but most insects have three pairs. Mites have eight legs, except for the initial developmental stage. Insects have six legs.
- Wings: Most adult insects have wings, but mites never do.
- Eyes: Most insects have compound eyes, but some have simple eyes. Mites lack compound eyes.
- Antennae: Spider mites lack antennae.

Mites are very common and can be found in many habitats. Spider mites can be red, brown, yellow, or green, depending on the species and the season. Mite bites can cause mild skin irritation and itching, and can be treated with antihistamines and topical corticosteroids.

Q#4 What do you understand by Riggs Diseases?

Question no08(2018)

Q#5 Name two pests of Cotton?

Here are some pests of cotton:

Aphids

The most common aphid pest in cotton, aphids cause feeding damage and transmit cotton bunchy top disease.

Thrips

One of the first pests to attack cotton crops, thrips attack small plants before true leaves develop.

Pink bollworm

This pest spoils the quality of cotton and is also called pink ballworm. An outbreak of pink bollworm usually starts after the fruit has buds and flowers.

American bollworm

This pest can cause losses to farmers. The young insect looks like a grub and is about the size of a match stick. It attacks and feeds on the inside of buds, flowers, and bolls.

Mites

Mites damage cotton by feeding on the sap of leaves, stems, and squares. This reduces the photosynthetic area and interferes with normal maturation of the cotton plant.

Q#6 Draw the diagram for the process of pearl formation in Molluscs?

See diagram on book (shukla & Upadhya)

Q#7 Give scientific names of Honey Bees?

Question no 05 (2018)

Q#8 Write problems of Silk Industry?

Question no 01(2019)

Q#9 Write down two pests of Household-Insects?

Common house insects After all, who likes seeing ants or flies infesting their home? Other common house insects include feared pests such as **termites**, **bed bugs** and **cockroaches**.

Q#10 Name two diseases of poultry?

Question no 03 (2019)

PAST PAPER (2021)

Q#1 Discuss spread of rats?

Rats can spread diseases to humans through direct contact, unsafe handling, or by eating or drinking contaminated food or water

The rat originated in southeast Asia, it turns out, spread to northeast Asia around **200,000** years ago, rested and spread to the Middle East around **3,600** years ago. Then, the rat evolutionary tree shows, about 2,600 years ago rats reached Africa

Q#2 Give products of piggery?

Pigs are raised for food and their skins. Pigs provide many products for humans, including:

- **Pork**
Bacon, ham, gammon, lard, pork rinds, salt pork, salo, guanciale, hog jowl, pancetta, and salami
- **Leather**
Used for clothing, shoes, handbags, sporting goods, and upholstery
- **Lard**
A fat used in cooking, shaving creams, soaps, make-up, baked goods, and other foods
- **By-products**
Organ meats like liver and heart, or tongue, brains, etc. These can end up in processed foods like sausage, or in pet food. Swine by-products are also important parts of products such as water filters, insulation, rubber, antifreeze, certain plastics, floor waxes, crayons, chalk, adhesives and fertilizer

Q#3 What is difference between mites and Ticks?

TICKS VERSUS MITES

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TICKS	MITES
Ticks are small arachnids that belong to the order Parasitiformes	Mites are arthropods that belong to the subclass Acari
Have a distinct body structure with a head, thorax, and abdomen	Have a more compact body without clear differentiation between body segments
Generally larger	Smaller in size
Have a relatively flat, oval-shaped body	Have a more compact and rounded body shape
Hypostome is visible and contains barbs pointing back toward the body	Hypostome is not visible and doesn't contain barbs
Very effective in anchoring to the host using hypostome	Easy to remove from the host
Cause diseases like Lyme disease, Rocky Mountain spotted fever, babesiosis, anaplasmosis, and tick-borne encephalitis	Cause disease conditions like scabies, chigger bites, allergies, and asthma

Q#4 Two pests of stored Grains?

Here are some pests that can affect stored grains:

- Rice weevil

A primary pest of stored grains, its larvae develop in sound kernels.

- Granary weevil

A black-brown weevil that can infest undamaged grain. Its larvae develop inside grain kernels, damaging wheat, corn, macaroni, oats, barley, sorghum, Kaffir seed, and buckwheat.

- Angoumois grain moth

A yellow-brown moth that attacks grains maturing in the field and in storage. Infested grains have a bad taste and smell.

Lesser grain borer

A dark brown, cylindrical beetle that is about 3mm long. Females lay up to 500 eggs in the grain.

Q#5 Two uses of honey?

Question no 04(2019)

Q#6 Give composition of a pearl ?

Pearls are made up of:

- Calcium carbonate: About **86%** of a pearl is made up of calcium carbonate (CaCO_3).
- Water: **2–4%** of a pearl is made up of water.

Conchiolin: About 10% of a pearl is made up of conchiolin, an organic binding agent.

Q#7 What is Stiffling?

Stiffling involves applying aggressive steps to the cocoons for hours or even days. These steps include:

- Sun exposure
- Hot steam from boiling water
- Hot air
- Heat drying
- Applying hot air
- Fumigation

After stiffling, the cocoons are boiled in water. This process helps loosen the adhesion of silk threads, making it easier to separate them.

Stiffling helps to store cocoons for a long time. Once the larvae comes out of the cocoon, a continuous thread of silk cannot be obtained from the cocoon

Q#8 Give difference between Male and Female Lac Insects?

Here are some differences between male and female lac insects:

- Size: Females are about three times larger than males.
- Wings: Males can be winged or wingless, while females are wingless. Winged males have one pair of translucent membranous forewings.

- Color: Males are pinkish red in color.
- Legs and antennae: Males have well-developed legs and antennae, while females have reduced legs and antennae.
- Mouthparts: Males have no mouthparts, while females have paurometabolous metamorphosis.
- Life cycle: Males are holometabolous, while females show paurometabolous metamorphosis.
- Morphology: Males are elongated, while females are round shaped.

Lac insects prefer warm climates and are distributed in tropical and sub-tropical regions. Female lac bugs secrete a resin called lac, which is used in both food and non-food industries.

Q#9 What do you know about IPM?

Integrated pest management (IPM) is a common-sense approach to controlling pests without using pesticides. It's an ecosystem-based strategy that uses a combination of techniques to prevent pests or their damage over time. IPM can be used in agricultural, urban, and natural areas.

IPM can help farmers:

- Create higher quality crops with less residues
- Increase farmer profitability
- Strengthen farmer knowledge
- Increase farmer knowledge of ecosystem functioning adapted to their local context

Q#10 Name two pests of Fruits?

Question no10 (2019)

PAST PAPER (2022)

Q#1 How Rabbits is used for research purpose?

Rabbits are used in many ways for research, including:

- Bioreactor: Rabbits are often used as bioreactors to produce antibodies. Rabbits are larger animals than other animals, so they can produce more antibodies without harming themselves. Rabbits are also genetically altered to produce antibodies that can be used in humans or to study human disease.
- Model for disease: Rabbits can be used as models for disease or biological systems. Rabbits can be used to study diseases and disorders of the eyes, skin, heart, and immune system, as well as cystic fibrosis, cancer, and spinal cord injury.
- Test animal: Rabbits can be used as "test animals" or sentinels for products.
- Safety testing: Rabbits are used in safety testing.
- Serology studies: Rabbits are used in serology studies because they produce serum antibodies in response to many different stimuli.

Studies on reproduction, aging, and physical development: Rabbits have a short lifespan, which makes them useful for studies on these topics.

Q#2 What is Swarming?

Swarming:

1. It is a process by which a queen leaves the old hive along with approximately one-half of the workers in that colony and takes a new shelter.
2. Swarming creates a vulnerable time in the life of every honey bee.
3. All bees tend to swarm near their hives, so if a swarm is visible, a nest is present nearby.

Q#3 Give four characteristics of Insecticide?

Here are some characteristics of insecticides:

- **Chemical nature**
Insecticides are classified into four groups based on their chemical nature: organic, synthetic, inorganic, and miscellaneous compounds.
- **Mode of action**
Insecticides can be classified into five groups based on their mode of action: physical poisons, protoplasmic poisons, respiratory poisons, nerve poisons, and poisons of a more general nature.
- **Target**
Organophosphates are a large class of contact insecticides that target the nervous system of insects. They disrupt nerve impulses by interfering with acetylcholinesterase enzymes and other cholinesterases.
- **Toxicity**
Organophosphates are effective at killing pests, but they can also be harmful to humans. They work by disrupting the nervous system, which can lead to seizures, respiratory problems, and even death.
- **Non-toxicity**
Insecticidal soap is made of detergent or ivory liquid and can be sprayed on plants to protect it from being eaten by insects. It is non-toxic and should not be sprayed in the Sunlight.

Q#4 What are main difference b/w Worker Bee and Drone?

Here are some differences between worker bees and drones:

- **Role:** Worker bees have many roles, including cleaning and repairing the hive, tending to the queen, and guarding the hive. Drones have only one role: to mate with virgin queens.
- **Sex:** Worker bees are female, while drones are male.
- **Stingers:** Worker bees have stingers, while drones do not.
- **Food:** Worker bees gather nectar and pollen, while drones do not. Drones also can't feed without help from worker bees.

- Eggs: Worker bees come from fertilized eggs, while drones come from unfertilized eggs. Drone eggs are also longer and wider than worker eggs.
- Lifespan: Worker bees live an average of 4–6 months, while drones live about 90 days.
- Mouth apparatus: Drones have a shorter mouth apparatus than worker bees, which prevents them from taking nectar from flowers

Q#5 What is Lac Culture?

Lac culture is the scientific management of lac insects to obtain a high amount of quality lac. This involves selection and maintenance of host plants, inoculation of host plants with healthy lac insects, collection and processing of lac and protection against enemies.

Q#6 Name two diseases of poultry?

Question no 10(2020)

Q#7 Differentiate b/w Insects and Pests?

Question no 06(2014)

Q#8 Differentiate b/w ticks & mites?

Question no 03(2021)

Q#9 What is IPM?

Question no 09(2021)

Q#10 Give Biological control of Nematodes?

Biological control agents (**BCAs**) can be used to control plant-parasitic nematodes. These BCAs include:

- Filamentous fungi: Such as the genus *Trichoderma*, mycorrhizal, and endophytic fungi
 - Bacteria: These are host-specific and can kill plant-parasitic nematodes
 - Actinomycetes: These are host-specific and can kill plant-parasitic nematodes
- Other methods for enhancing natural biological control include:**
- Chitin amendments: These stimulate chitinolytic organisms that can degrade nematode eggshells
 - Collagen amendments: These stimulate collagenolytic organisms that can degrade the cuticle of vermiform nematodes

Plant-parasitic nematodes are a major threat to agricultural production worldwide. Some say that conserving soil biodiversity is crucial to enhance the reliability of biological crop protection against soil-borne pests and diseases.

Q#11 What is Antifeedant?

Antifeedants are natural or synthetic substances that prevent or inhibit pests, particularly insects, from

feeding. They are chemical compounds that deter or inhibit the feeding of an organism.

Antifeedants are organic compounds produced by plants to repel herbivores. They are secondary metabolites, meaning they are not essential for the plant's metabolism.

Some examples of antifeedants include:

Garlic, Rotenone, Veratridines, Pyrethrins, Nicotine, Green garlic, Tobacco plants.

Most antifeedants act directly on the taste system of insects. Some antifeedants act through antagonizing the action of the γ -aminobutyric acid (GABA) on insect neurons to induce feeding deterrence

Q#12 Differentiate B/w Musk & Ivory?

Musk is a synthetic ingredient used in perfumery, while ivory is a hard, white substance that comes from the tusks of elephants and other animals.

Musk is a large molecule that can last for hours on the skin and can make a fragrance smell warm, intimate, and sensual. It can also tie all the ingredients together and make the fragrance last longer. Musk can have a variety of scents, including clean and fresh, earthy and woody, sweet and natural, and light and woody.

Ivory is a hard, white substance that comes from the tusks of elephants, hippopotami, walruses, warthogs, sperm whales, and narwhals. It is prized for its close-grained texture, adhesive hardness, mellow color, and pleasing smoothness. Ivory can be painted or bleached, and is an excellent material for carving.

Q#14 Give biological control of sand Fly?

Here are some biological control methods for sand flies:

- Mites and spiders
These predators act as biological control agents for the larvae and adults of sand flies.
- Microbial strains
The presence of microbial strains in both larvae and adult sand flies can help control sand fly populations.
- Bacillus sphaericus
In Israel, Bacillus sphaericus is applied to sugar solutions near sand fly habitats to reduce the adult population and kill larvae.
- Free-living fungi

Peter Ngunjiri of Daystar University in Kenya is developing a biological control for sand flies using fungi found in the local soil.

- Toxic sugar baits

A five-week trial period showed that sand fly populations were reduced by 76.9% at sugar poor sites and 83.2% at sugar rich sites when ATSB is presented as a bait station or applied to vegetation.

Q#15 Write the name of species for cotton pests?

PESTS OF COTTON :: Major Pests :: **Whitefly**

Severe infestation results in premature defoliation, development of sooty mould, shedding of buds and bolls and poor boll opening. It also transmits the leaf curl virus disease of cotton.

Jassid. Amrasca biguttula is one of the most damaging pests of cotton crop in Pakistan. Damage of Cotton Jassid is very different as compared to that of other insect pests of cotton. Cup formation of leaves is observed in the peak of its attack.

PAST PAPER(2023)

Q#1 Why are pesticides considered to be hazardous?

Pesticides can be hazardous for many reasons, including:

- Acute toxicity
A pesticide's ability to cause injury to a person or animal from a single exposure, generally of short duration. Pesticides can be toxic through dermal (skin), inhalation (lungs), oral (mouth), and eyes.
- Chronic exposure
Can affect the skin, eyes, nervous system, cardiovascular system, gastrointestinal tract, liver, kidneys, reproductive system, endocrine system, immune system and blood.
- Endocrine disruption
Can cause disruption of cellular signaling by binding to hormone receptors present on cell membrane or nucleus.
- Environmental impact
Can contaminate soil, water, turf, and other vegetation, drive biodiversity loss, and lead to pest resistance.
- Toxicity to other organisms
Can be toxic to a host of other organisms including birds, fish, beneficial insects, and non-target plants.
- Soil effects
Can kill bacteria and other organisms beneficial to the soil, increase nitrate content in the soil, alter pH levels, and bioaccumulate.

Q#2 Give composition of a pearl?

Just like the shell of a mollusk, a pearl is composed of calcium carbonate (mainly aragonite or a mixture of aragonite and calcite) in minute crystalline form, which has deposited in concentric layers. ~ 86 % calcium carbonate (CaCO₃) 2 - 4 % water. ~ 10 % conchiolin (an organic binding agent)

Q#3 Write problems of silkworm?

Major silkworm diseases are viral such as nuclear polyhedrosis (*Bombyx mori* nucleopolyhedrovirus; BmNPV), cytoplasmic polyhedrosis (*Bombyx mori* cytoplasmic polyhedrovirus; BmCPV), infectious flacherie (*Bombyx mori* infectious flacherie virus; BmIFV), densovirus (*Bombyx mori* densovirus; BmDENV)

Q#4 Give products of piggery?

Question no2(2021)

Q#5 Diseases spread by rats?

Rats can carry many diseases, including:

- **Hantavirus**

A viral disease found in wild rodents, such as rats and mice. Hantaviruses found in North America can cause Hantavirus Pulmonary Syndrome, a severe lung disease that can be fatal.

- **Leptospirosis**

Also known as Weil's disease, this disease is contracted from the urine of infected rats. The bacteria get into your body through cuts and scratches or through the lining of the mouth, throat and eyes after contact with infected urine or contaminated water.

- **Rat bite fever**

An infectious disease caused by two different bacteria:

- **Streptobacillary RBF:** Caused by *Streptobacillus moniliformis*, the only reported bacteria that causes RBF in North America
- **Spirillary RBF:** Also known as sodoku, caused by *Spirillum minus*, common in Asia

Q#6 What is dagger nematodes?

Dagger nematodes, also known as the genus *Xiphinema*, are plant parasites that can cause economic damage and death. They are widely distributed in temperate and tropical areas, including South America, North America, Europe, Asia, Australia, New Zealand, and Africa.

Dagger nematodes are ectoparasitic, which means that all stages, except eggs, attack and feed on the roots of the host plants. The nematode inserts its long stylet deep into the root while the body remains outside the root, in the soil. The stylet punctures cell walls as it penetrates plant tissues.

Dagger nematodes have six life cycle stages and have the following characteristics:

- Long body (2 to 6 mm)
- Flat and smooth lip region
- Parthenogenesis, a form of reproduction that does not require males

Compared to many other species of plant-parasitic nematodes, they have long life cycles but fortunately do not produce large numbers of eggs (less than 50).

Dagger nematodes are economically important on grape, strawberry, hops and a few other crops. Major species include *X. americanum*, *X. diversicaudatum*.

Q#7 Two important edible freshwater fishes?

Some important edible freshwater fish include:

- **Tilapia**

A popular fish that lives in shallow water, lakes, and rivers, and is also found in brackish water. Tilapia is one of the most important species in aquaculture, after salmon and carp.

- **Trout**

A nutritious fish that is rich in omega-3 fatty acids and protein.

- **Catfish**

Depending on where you live, catfish may be the most widely eaten freshwater fish.

Q#8 Three castes of Honeybees?

Question no 06 (2019)

Q#9 What are the two products of Apiculture?

Question no 07(2019)

Q#10 What is Swarming?

Question no 2 (2022)

Q#11 What are two important diseases of poultry?

Question no 03(2016)

Q#12 What is antifeedant?

Question no 11(2022)

Q#13 Differentiate b/w Musk & ivory?

Question no 12(2022)

Q#14 What is Lac culture?

Question no5(2022)

Q#15 Differentiate b/w Ticks & Mites?

Question no 08(2022)

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